

## BREATHEBOX · REQUIREMENTS

# BreatheBox requirements

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## DOCUMENT CONTROL

|             |                                         |         |                |
|-------------|-----------------------------------------|---------|----------------|
| Document ID | BBX-REQ-001                             | Version | 0.3            |
| Status      | <b>Draft</b>                            | Date    | 2026-09-25     |
| Author      | Amish Chadha                            | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept           |         |                |
| Repository  | github.com/BoujeeEnjinia1701/breathebox |         |                |
| Source      | docs/03-requirements.md @ 1cf85f1       |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                                         |
|---------|------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.3     | 2026-09-25 | Amish Chadha | Status against BBX-CAL-001 at TRL 3; R7 condensate figure corrected to 0.21 L/h; R8 and R15 notes per BBX-DDR-001 (adopted for TRL 3, open for Amish's review) |
| 0.2     | 2026-09-25 | Amish Chadha | First measurable requirements for TRL 2, with status against the concept                                                                                       |
| 0.1     | 2026-09-25 | Amish Chadha | Initial scaffold                                                                                                                                               |

## BreatheBox requirements

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These requirements are proposals for review, not yet validated with users, and will be revised after co-design (see BBX-PRB-001). Status is judged against the TRL 3 calculations in BBX-CAL-001 and the parametric model in `cad/src/model.py` ; nothing has been built or measured. No target was relaxed at TRL 3. The R7 condensate figure was corrected from 0.15 to 0.21 L/h because the calculation showed a higher worst case.

The **design case** is a 30 m<sup>3</sup> bedroom with two sleeping adults, 20 °C and 50 % relative humidity indoors, 0 °C outdoors, and a vertical sliding sash window with a 900 mm clear width.

Table 1. Requirements and status at TRL 3.

| ID  | Requirement                              | Target                                                                                                                                                   | Status at TRL 3 (BBX-CAL-001)                                                                                              | Verification (TRL 4 or later)     |
|-----|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| R1  | Balanced ventilation over a useful range | Supply and exhaust each adjustable from 30 to 70 m <sup>3</sup> /h (18 to 41 cfm); supply and exhaust within 10 % of each other outside frost mode       | Met on paper: 80 m <sup>3</sup> /h per stream at full speed with loaded filters; balance by fan tach                       | Flow hood measurement             |
| R2  | Recover heat from the exhaust air        | Sensible effectiveness 75 % or more at 50 m <sup>3</sup> /h and balanced flow                                                                            | Met on paper: 79.8 %                                                                                                       | Temperatures on the four ports    |
| R3  | Low running power                        | Fans and controls 12 W or less at 50 m <sup>3</sup> /h                                                                                                   | Met on paper: 5.6 W clean, 7.4 W with loaded filters                                                                       | Power meter                       |
| R4  | Keep bedroom air fresh                   | CO <sub>2</sub> at or below 1,000 ppm (24-hour average) in the design case, following the Health Canada residential guideline                            | Met on paper: 607 ppm 24 h mean at 50 m <sup>3</sup> /h; 980 ppm overnight                                                 | CO <sub>2</sub> logging           |
| R5  | Filter the supply air                    | ISO 16890 ePM1 50 % or better on supply; coarse filter on exhaust to protect the core                                                                    | Met by specification                                                                                                       | Filter data sheet                 |
| R6  | Quiet at night                           | 30 dB(A) or less at 1 m from the room face at 50 m <sup>3</sup> /h                                                                                       | <b>Not met:</b> about 39 dB(A) (estimate); 30 dB(A) near 32 m <sup>3</sup> /h                                              | Sound meter                       |
| R7  | Handle condensate and frost              | All condensate drains outdoors (up to about 0.21 L/h); runs without core blockage down to -10 °C outdoor, using a frost mode that may reduce supply flow | Met on paper: frost onset about -2.2 °C; at -10 °C the supply runs at 33 % of the exhaust                                  | Cold-chamber or winter test       |
| R8  | Fit common windows                       | Vertical sliding sash windows with 700 to 1,000 mm clear width and sill 150 mm or deeper; adapter panel for horizontal sliders                           | Met on paper for sash: the hoods need 686 mm. Casement and tilt-and-turn windows are a later variant (BBX-DDR-001 item A2) | Fit check against a window survey |
| R9  | Renter-friendly installation             | No drilling of frame or wall; one person installs or removes in 30 min or less; unit mass 12 kg or less                                                  | Not verifiable at TRL 3: mass 10.5 kg met on paper; install time needs a trial                                             | Timed trials                      |
| R10 | Electrical safety                        | Only 24 V SELV inside the unit, from a certified plug-in adapter; fused low-voltage input; no user-accessible mains                                      | Met by design: 36 W adapter, 2 A fuse; worst case 35.4 W                                                                   | Design review                     |
| R11 | Weather and outdoor air                  | Rain-proof downward-facing hoods; insect mesh 1.5 mm or finer; 400 mm or more between supply intake and exhaust outlet                                   | Met on paper: 420 mm; 1 mm mesh                                                                                            | Spray test                        |
| R12 | Security and child safety                | Raised sash lockable onto the insert panel; no opening wider than 100 mm through the installed unit                                                      | Met by design, unverified: largest opening 2.35 mm                                                                         | Design review with a locking bar  |
| R13 | Serviceable                              | Filters changed without tools in 2 min or less; core removable for washing; all parts replaceable with a screwdriver                                     | Not verifiable at TRL 3: lift-off lid and slide-out filters and core in the model                                          | Timed trials                      |

| ID  | Requirement     | Target                                                                                                          | Status at TRL 3 (BBX-CAL-001)                                                                                                   | Verification (TRL 4 or later) |
|-----|-----------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| R14 | Local data only | CO2, humidity and temperature shown on the unit; works with no network or cloud account; optional local logging | Not verifiable at TRL 3: no firmware yet                                                                                        | Firmware review               |
| R15 | Affordable      | Prototype parts \$220 or less                                                                                   | <b>Not met:</b> \$255 (priced BOM). Also \$5 over the \$250 recommended at TRL 2, which is awaiting Amish (BBX-DDR-001 item A1) | Priced BOM ( bom/bom.csv )    |

## ■ Requirements not met

- **R6 (noise) not met:** about 39 dB(A) at 1 m at 50 m<sup>3</sup>/h against 30 dB(A). Options are in docs/REVIEW.md , awaiting Amish.
- **R15 (cost) not met:** \$255 against \$220 in project.yaml , and \$5 over the proposed \$250.
- **Not verifiable at TRL 3:** R9 (install time), R13 (filter change time) and R14 (firmware).

## ■ Assumptions

- Air density 1.2 kg/m<sup>3</sup> and specific heat 1,005 J/(kg·K).
- CO2 generation about 0.014 m<sup>3</sup>/h per sleeping adult (estimate); outdoor CO2 about 420 ppm.
- Core, filter, fan and noise values are class values stated in BBX-CAL-001 section 1, not datasheet values.